optimisation problems. It combines quantum circuits with classical optimisation steps to solve problems that involve finding optimal arrangements, such as job scheduling and network optimisation.

These algorithms can be implemented using open source tools like PennyLane and TensorFlow Quantum, both of which support hybrid models and provide prebuilt functions for VQE and QAOA. A practical example would be using VQE in a hybrid model to optimise a neural network's weights, potentially leading to faster convergence and better model accuracy.

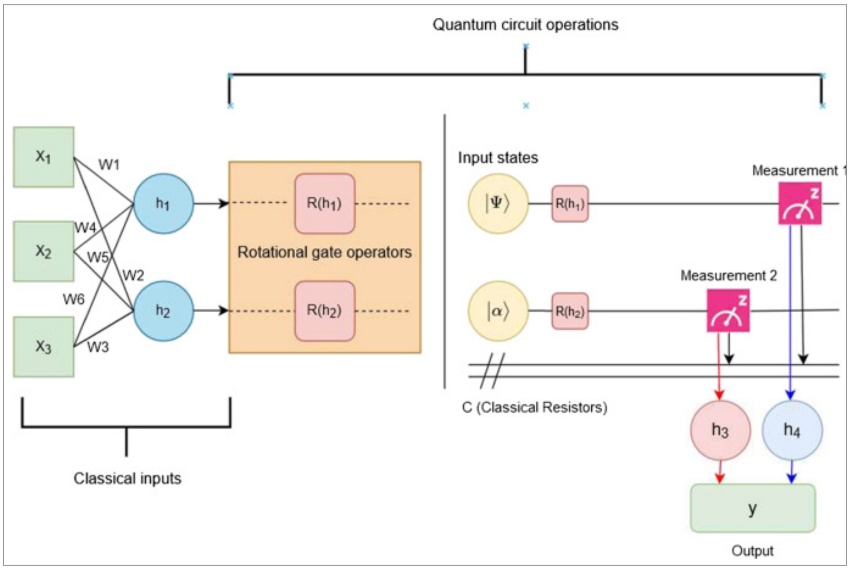

Figure 3: Hybrid quantum neural networks

Challenges and future directions

Quantum machine learning holds significant promise, but there are challenges that need to be addressed for it to achieve widespread adoption. One of the biggest obstacles is quantum hardware limitations. Current quantum computers are still in the noisy intermediate-scale quantum (NISQ) era, meaning that they are prone to errors and lack scalability. These limitations affect the reliability of quantum machine learning models and hinder their ability to generalise to larger datasets.

Scalability is another concern, as quantum machine learning models are difficult to scale due to hardware constraints and algorithmic challenges. Moreover, developing models that can generalise across different data distributions requires advances in both quantum hardware and quantum algorithms. There is a pressing need for error correction techniques that can reduce noise in quantum computations as well as for new algorithms that are robust and can address quantum-specific limitations.

Despite these challenges, the future of quantum-enhanced machine learning is promising. With ongoing advancements in quantum hardware and open source software, we can expect to see more practical applications of quantum machine learning in areas such as healthcare, finance, and materials science. Emerging trends, such as quantum federated learning and distributed quantum AI, are paving the way for new possibilities in data privacy and collaborative machine learning. The integration of quantum and classical systems is also likely to grow, offering powerful hybrid solutions for complex problems.

Open source frameworks are accelerating the development of quantum-enhanced machine learning, bridging the gap between quantum computing and AI. By offering accessible and cross-platform tools, frameworks like Qiskit, PennyLane, and TensorFlow Quantum enable researchers and developers to experiment, innovate, and share their work in this transformative field. While there are still challenges to overcome, the future of quantum machine learning is bright. As quantum hardware improves and new algorithms are developed, we can expect quantum-enhanced AI models to tackle increasingly complex tasks, opening new frontiers in both quantum computing and artificial intelligence. END

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